Symbiosis Institute of Technology, Pune

Faculty of Engineering
CSE- Academic Year 2026-27
Java Programming – Lab Batch 2024-28

Lab Assignment No:- 30	
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Batch	A1
Class	CSE
Academic Year & Semester	2026-2027/3RD
Date of Submission	18/10/2026
Title of Assignment:	Write a JSP application to demonstrate CRUD operations.
Theory:	CRUD stands for Create, Read, Update and Delete, the four basic operations performed on stored data. In a JSP application the user fills in an HTML form, and the JSP page uses JDBC to run the matching SQL statement on a MySQL database: INSERT creates a record, SELECT reads records, UPDATE changes an existing record and DELETE removes one. PreparedStatement is used so that user input is passed to the query as parameters, which avoids SQL injection and handles data types correctly. Exceptions are caught and shown on the page so that wrong input or a failed connection does not break the application. CRUD is the basis of employee, hospital, banking and shopping systems.
Practice Questions	1. Create a JSP-based application to add, update, delete, and view employee details.

CODE:

```
-- MySQL setup (run once)
CREATE DATABASE jsp_lab;
USE jsp_lab;
CREATE TABLE employee (
    id INT PRIMARY KEY, name VARCHAR(50),
    department VARCHAR(50), salary DOUBLE);
CREATE TABLE patient (
    patient_id INT PRIMARY KEY, name VARCHAR(50), age INT,
    disease VARCHAR(50), treatment_status VARCHAR(50));

-- employee.jsp
<%@ page import="java.sql.*" %>
<%
    String vId = request.getParameter("id");
    String vName = request.getParameter("name");
    String vDept = request.getParameter("department");
    String vSal = request.getParameter("salary");
    if (vId == null) vId = "";
    if (vName == null) vName = "";
    if (vDept == null) vDept = "";
    if (vSal == null) vSal = "";
%>
<!DOCTYPE html>
<html>
<head>
    <title>Employee CRUD</title>
</head>
<body>
    <h2>Employee Management System</h2>
    <form method="post">
        Employee ID:
        <input type="number" name="id" value="<%= vId %>"
required>
        <br><br>
        Name:
        <input type="text" name="name" value="<%= vName %>">
        <br><br>
        Department:
        <input type="text" name="department" value="<%= vDept %>">
```

```
        <br><br>
        Salary:
        <input type="number" step="0.01" name="salary"
               value="<%= vSal %>">
        <br><br>
        <input type="submit" name="action" value="Add">
        <input type="submit" name="action" value="Update">
        <input type="submit" name="action" value="Delete">
        <input type="submit" name="action" value="View">
    </form>
    <hr>
<%
    String url = "jdbc:mysql://localhost:3306/jsp_lab";
    String dbUser = "root";
    String dbPass = "your_mysql_password";
    String action = request.getParameter("action");

    if (action != null) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                url, dbUser, dbPass);
            PreparedStatement ps;

            String idStr = request.getParameter("id");
            String name = request.getParameter("name");
            String dept = request.getParameter("department");
            String salStr = request.getParameter("salary");

            if (action.equals("Add")) {
                ps = con.prepareStatement(
                    "INSERT INTO employee VALUES (?, ?, ?, ?)");
                ps.setInt(1, Integer.parseInt(idStr));
                ps.setString(2, name);
                ps.setString(3, dept);
                ps.setDouble(4, Double.parseDouble(salStr));
                ps.executeUpdate();
                out.println("<p>Employee added.</p>");

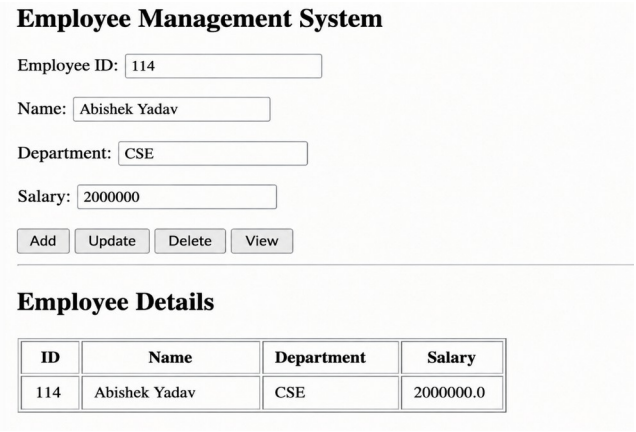
            } else if (action.equals("Update")) {
                ps = con.prepareStatement(
                    "UPDATE employee SET name=?, department=?, "
                    + "salary=? WHERE id=?");
                ps.setString(1, name);
                ps.setString(2, dept);
                ps.setDouble(3, Double.parseDouble(salStr));
                ps.setInt(4, Integer.parseInt(idStr));
                int rows = ps.executeUpdate();
                out.println("<p>" + rows
                    + " record(s) updated.</p>");

            } else if (action.equals("Delete")) {
                ps = con.prepareStatement(
                    "DELETE FROM employee WHERE id=?");
                ps.setInt(1, Integer.parseInt(idStr));
                int rows = ps.executeUpdate();
                out.println("<p>" + rows
                    + " record(s) deleted.</p>");

            } else if (action.equals("View")) {
                Statement st = con.createStatement();
                ResultSet rs = st.executeQuery(
                    "SELECT * FROM employee");
                out.println("<h3>Employee Details</h3>");
                out.println("<table border='1' cellpadding='8'>");
                out.println("<tr><th>ID</th><th>Name</th>"
                    + "<th>Department</th><th>Salary</th></tr>");
                while (rs.next()) {
                    out.println("<tr><td>" + rs.getInt("id")
                        + "</td><td>" + rs.getString("name")
                        + "</td><td>" + rs.getString("department")
                        + "</td><td>" + rs.getDouble("salary")
                        + "</td></tr>");
                }
                out.println("</table>");
            }
            con.close();
        } catch (Exception e) {
            out.println("<p>Error: " + e.getMessage() + "</p>");
        }
    }
%>
</body>
```

```
</html>
```

OUTPUT:



QUESTION	Create a JSP-based application to manage patient records using CRUD operations. Include fields such as patient ID, name, age, disease, and treatment status.

CODE:

```
-- patient.jsp
<%@ page import="java.sql.*" %>
<%
    String vId = request.getParameter("patient_id");
    String vName = request.getParameter("name");
    String vAge = request.getParameter("age");
    String vDis = request.getParameter("disease");
    String vStat = request.getParameter("treatment_status");
    if (vId == null) vId = "";
    if (vName == null) vName = "";
    if (vAge == null) vAge = "";
    if (vDis == null) vDis = "";
    if (vStat == null) vStat = "";
%>
<!DOCTYPE html>
<html>
<head>
    <title>Patient CRUD</title>
</head>
<body>
    <h2>Patient Management System</h2>
    <form method="post">
        Patient ID:
        <input type="number" name="patient_id"
               value="<%= vId %>" required>
        <br><br>
        Name:
        <input type="text" name="name" value="<%= vName %>">
        <br><br>
        Age:
        <input type="number" name="age" value="<%= vAge %>">
        <br><br>
        Disease:
        <input type="text" name="disease" value="<%= vDis %>">
        <br><br>
        Treatment Status:
        <input type="text" name="treatment_status"
               value="<%= vStat %>">
        <br><br>
        <input type="submit" name="action" value="Add">
        <input type="submit" name="action" value="Update">
        <input type="submit" name="action" value="Delete">
        <input type="submit" name="action" value="View">
    </form>
    <hr>
<%
    String url = "jdbc:mysql://localhost:3306/jsp_lab";
    String dbUser = "root";
    String dbPass = "your_mysql_password";
    String action = request.getParameter("action");

    if (action != null) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection con = DriverManager.getConnection(
                url, dbUser, dbPass);
            PreparedStatement ps;

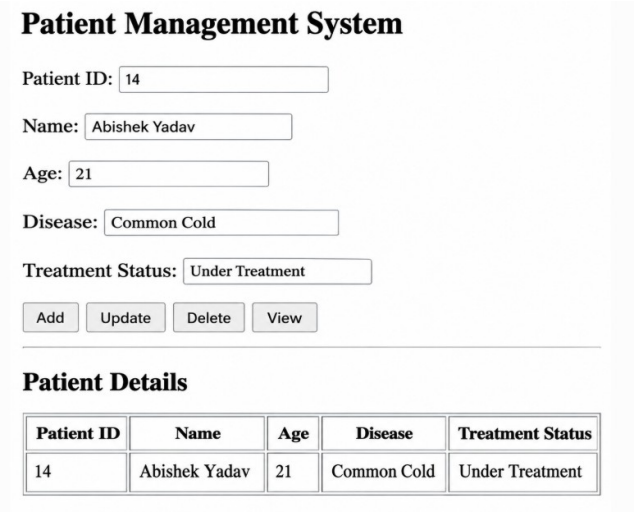
            String idStr = request.getParameter("patient_id");
            String name = request.getParameter("name");
            String ageStr = request.getParameter("age");
            String disease = request.getParameter("disease");
            String status =
                request.getParameter("treatment_status");

            if (action.equals("Add")) {
                ps = con.prepareStatement(
                    "INSERT INTO patient VALUES (?, ?, ?, ?, ?)");
                ps.setInt(1, Integer.parseInt(idStr));
                ps.setString(2, name);
                ps.setInt(3, Integer.parseInt(ageStr));
                ps.setString(4, disease);
                ps.setString(5, status);
                ps.executeUpdate();
                out.println("<p>Patient added.</p>");

            } else if (action.equals("Update")) {
                ps = con.prepareStatement(
                    "UPDATE patient SET name=?, age=?, "
                    + "disease=?, treatment_status=? "
                    + "WHERE patient_id=?");
                ps.setString(1, name);
                ps.setInt(2, Integer.parseInt(ageStr));
                ps.setString(3, disease);
                ps.setString(4, status);
                ps.setInt(5, Integer.parseInt(idStr));
                int rows = ps.executeUpdate();
                out.println("<p>" + rows
                    + " record(s) updated.</p>");

            } else if (action.equals("Delete")) {
                ps = con.prepareStatement(
                    "DELETE FROM patient WHERE patient_id=?");
                ps.setInt(1, Integer.parseInt(idStr));
                int rows = ps.executeUpdate();
                out.println("<p>" + rows
                    + " record(s) deleted.</p>");

            } else if (action.equals("View")) {
                Statement st = con.createStatement();
                ResultSet rs = st.executeQuery(
                    "SELECT * FROM patient");
                out.println("<h3>Patient Details</h3>");
                out.println("<table border='1' cellpadding='8'>");
                out.println("<tr><th>Patient ID</th><th>Name</th>"
                    + "<th>Age</th><th>Disease</th>"
                    + "<th>Treatment Status</th></tr>");
                while (rs.next()) {
                    out.println("<tr><td>"
                        + rs.getInt("patient_id")
                        + "</td><td>" + rs.getString("name")
                        + "</td><td>" + rs.getInt("age")
                        + "</td><td>" + rs.getString("disease")
                        + "</td><td>"
                        + rs.getString("treatment_status")
                        + "</td></tr>");
                }
                out.println("</table>");
            }
            con.close();
        } catch (Exception e) {
            out.println("<p>Error: " + e.getMessage() + "</p>");
        }
    }
%>
</body>
</html>
```

OUTPUT:	

Conclusion	The JSP applications successfully demonstrate CRUD operations. Employee and patient records were added, updated, deleted and displayed using JSP pages, JDBC PreparedStatements and a MySQL database, which shows how a web application interacts with a database to manage records.